



BAULKHAM HILLS HIGH SCHOOL

2017
YEAR 12
TASK 3 JUNE

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- All relevant mathematical reasoning and/or calculations must be shown
- A standard integral sheet is provided at the back of this paper.

Total marks – 39
(Pages 2-5)

Questions 1 to 3
(13 marks each)

QUESTION 1 (13 marks) Start on the appropriate page of your answer booklet

(a) Find $\int \frac{1}{x^2 + 2x + 5} dx$ **2**

(b) Find the value of the constants A and B such that

(i) $\frac{1}{u^2 + 3u + 2} = \frac{A}{1 + u} + \frac{B}{2 + u}$ **2**

(ii) Hence deduce the exact value of the integral $\int_0^1 \frac{e^t}{e^{2t} + 3e^t + 2} dt$ **3**

(c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + 3\sin x + 2\cos x}$ using the substitution $t = \tan \frac{x}{2}$. **3**

(d) By using the fact that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$. **3**

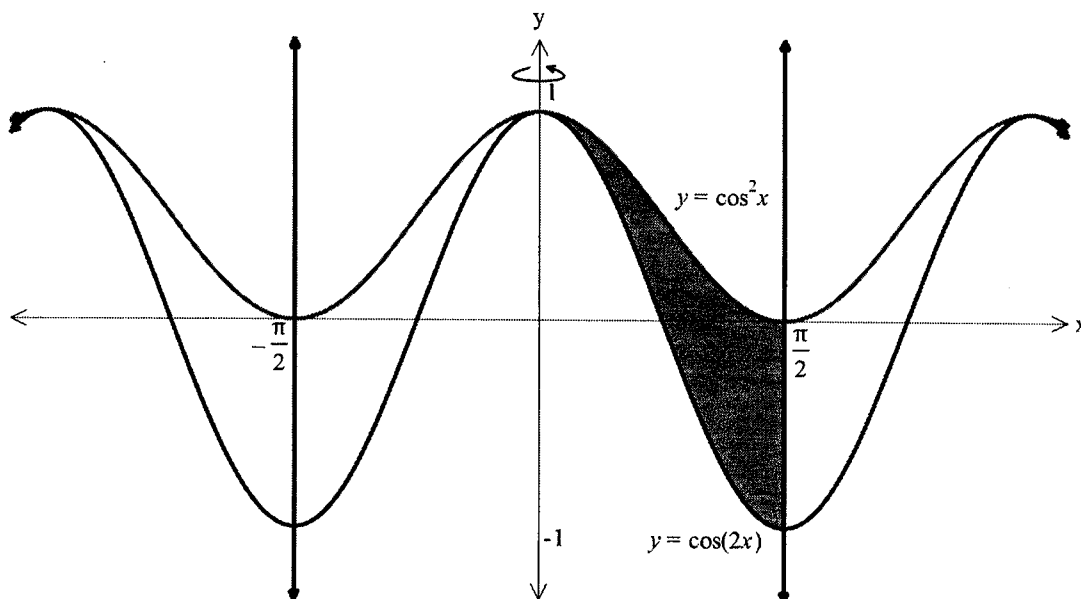
End of Question 1

QUESTION 2 (13 marks) Start on the appropriate page of your answer booklet.

(a) Find $\int \frac{x}{\sqrt{16-x^2}} dx$

2

- (b) The area between the graphs of $y = \cos^2 x$ and $y = \cos 2x$ for $0 \leq x \leq \frac{\pi}{2}$ and $x = \frac{\pi}{2}$ is rotated about the y axis.



By considering cylindrical shells, find the volume of the solid formed, in terms of π .

4

(c) Use the properties of odd and even functions to evaluate $\int_{-4}^4 \cos x (e^x - e^{-x}) dx$.

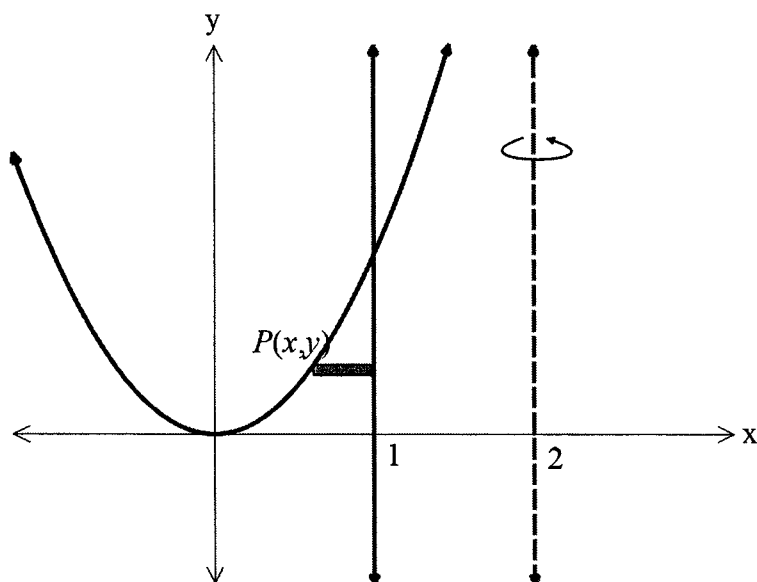
3

Justify your answer with reasoning.

Question 2 continues on the next page

- (d) The area bounded by $x = 1$, $y = 0$ and $y = x^2$ is rotated about the line $x = 2$.

The volume of the solid formed is to be determined by taking slices perpendicular to the axis of rotation.



- | | | |
|------|---|----------|
| (i) | Show that the area of the annulus for a slice is $A = \pi(3 - 4\sqrt{y} + y)$ | 1 |
| (ii) | Find the volume of the solid formed. | 3 |

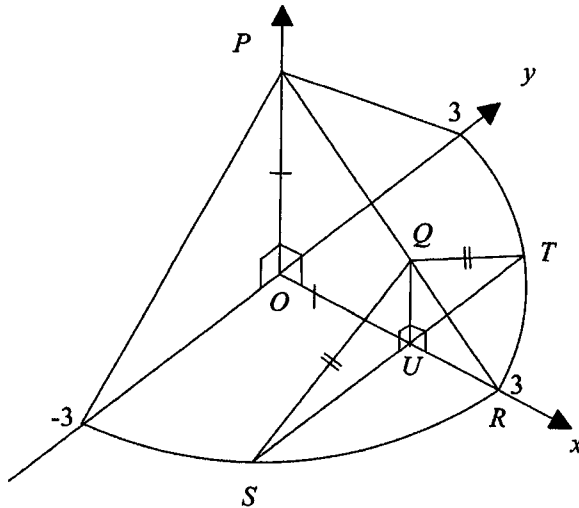
End of Question 2

QUESTION 3 (13 marks) Start on the appropriate page of your answer booklet.

(a) Find $\int x^3 \sin x^2 dx$

3

- (b) A solid figure has a semi circular base of radius 3cm. Cross sections taken perpendicular to the x axis are isosceles triangles.



The vertical cross section containing the radius OR of the base of the solid is an isosceles right angled triangle ORP where $OR = OP$.

- (i) Show that the area of the triangle SQT is given by

2

$$A = (3 - x)(9 - x^2)^{\frac{1}{2}} \text{ where } x = OU.$$

- (ii) Show that the volume of the solid is $\frac{1}{4}(27\pi - 36) \text{ cm}^3$.

3

(c) Let $I_n = \int \frac{x^n}{1 + x^2} dx$ for $n \geq 0$.

(i) Show that $I_n = \frac{x^{n-1}}{n-1} - I_{n-2}$

2

(ii) Hence evaluate $\int_1^3 \frac{x^5}{1 + x^2} dx$.

3

End of examination

EXTENSION 2 - JUNE 2017

$$1 \ a) \int \frac{dx}{x^2 + 2x + 5} = \int \frac{dx}{(x+1)^2 + 4}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$$

- ② correct answer
- ① completes the square

$$b) (i) \frac{1}{u^2 + 3u + 2} = \frac{A}{1u} + \frac{B}{2u}$$

$$1 = A(2u) + B(1u)$$

$$\text{let } u = -2, 1 = 0 - B$$

$$B = -1$$

$$\text{let } u = -1, 1 = A + 0$$

$$\therefore A = 1 \text{ and } B = -1$$

$$(ii) \int \frac{e^t dt}{e^{2t} + e^t + 2} \quad \text{let } u = e^t$$

$$du = e^t dt$$

$$\text{when } t=1, u=2$$

$$t=0, u=1$$

$$\int \frac{du}{u^2 + 3u + 2}$$

$$= \int \frac{du}{u+1} - \int \frac{du}{u+2}$$

$$= \left[\ln(u+1) - \ln(u+2) \right]_1^2$$

$$= \left[\ln \left(\frac{u+1}{u+2} \right) \right]_1^2$$

$$= \ln \left(\frac{e+1}{e+2} \right) - \ln \left(\frac{2}{3} \right)$$

$$= \ln \frac{3(e+1)}{2(e+2)}$$

accept either

- ③ correct answer
- ② converts to integral in u and integrates
- ② converts to integral in u , incorrectly integrates and evaluates
- ① converts to integral in u

$$1 \ (c) \int_0^{\frac{\pi}{2}} \frac{dx}{2 + 3 \sin x + 4 \cos x} \quad t = \tan \frac{x}{2}$$

$$\frac{2dt}{1+t^2} = dx$$

$$\text{when } x = \frac{\pi}{2}, t = \tan \frac{\pi}{4} = 1$$

$$x = 0, t = \tan 0 = 0$$

$$\int_0^1 \frac{2dt}{2 + 3 \frac{2t}{1+t^2} + 2 \left(\frac{1-t^2}{1+t^2} \right)}$$

$$= \int_0^1 \frac{2dt}{2 + 6t + 2 - 2t^2}$$

$$= \int_0^1 \frac{2dt}{6t + 4}$$

$$= \int_0^1 \frac{dt}{3t + 2}$$

$$= \frac{1}{3} \left[\ln(t+2) \right]_0^1$$

$$= \frac{1}{3} \ln 5 - \ln 2$$

$$= \frac{1}{3} \ln \left(\frac{5}{2} \right)$$

- ③ correct answer
- ② expresses as integral in t and correctly integrates
- ② expresses as integral in t and evaluates integral with incorrect limits
- ① expresses as integral in t and integrates with wrong limits

1 (d) next page

$$\begin{aligned}
 1. (d) \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\
 = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \quad \checkmark \\
 = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \\
 = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx
 \end{aligned}$$

- ① correct answer
 ② applies formula and evaluates $\left[\tan^{-1}(\cos x) \right]_0^{\pi}$
 ③ applies formula and obtains expression for $2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$
 ④ applies formula

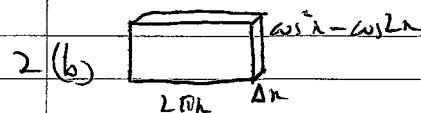
$$\text{So that } \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^{\pi} \left(\frac{\pi \sin x}{1 + \cos^2 x} - \frac{x \sin x}{1 + \cos^2 x} \right) dx$$

$$\begin{aligned}
 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \quad \checkmark \\
 &= -\pi \left[\tan^{-1}(\cos x) \right]_0^{\pi} \\
 &= -\pi \left(\tan^{-1}(\cos \pi) - \tan^{-1}(\cos 0) \right) \quad \checkmark \\
 &= -\pi \left(\tan^{-1}(-1) - \tan^{-1}(1) \right) \\
 &= -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi^2}{2} \\
 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi^2}{4} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 2. (a) \int \frac{x dx}{\sqrt{16 - x^2}} \\
 = \int -2x (16 - x^2)^{-\frac{1}{2}} dx \\
 = -\frac{1}{2} (16 - x^2)^{\frac{1}{2}} + C \\
 = -\sqrt{16 - x^2} + C
 \end{aligned}$$

- ② correct answer
 ① obtains a $\sqrt{16 - x^2}$ where $a = -k$ (or equivalent merit)



2. (b)

$$\begin{aligned}
 \text{Volume of shell } \Delta V &= 2\pi x (\cos x - \cos 2x) \Delta x \\
 &= \pi x (2\cos x - 2\cos 2x) \Delta x \\
 &= \pi x (\cos 2x + 1 - 2\cos 2x) \Delta x \\
 &= \pi x (1 - \cos 2x) \Delta x
 \end{aligned}$$

- ④ correct answer
 ③ integrates by parts
 ② expresses volume integral as $\pi \int x - x \cos 2x dx$
 ① finds volume of a slice

$$\begin{aligned}
 V &= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\frac{\pi}{2}} \pi (x - x \cos 2x) \Delta x \\
 &= \pi \int_0^{\frac{\pi}{2}} x - x \cos 2x dx \\
 &= \pi \left[\frac{x^2}{2} \right]_0^{\frac{\pi}{2}} - \pi \int_0^{\frac{\pi}{2}} x \cos 2x dx \\
 &= \pi \frac{\pi^2}{8} - \pi \left[\left[\frac{x \sin 2x}{2} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \frac{\sin 2x}{2} dx \right] \\
 &= \frac{\pi^3}{8} - \pi \left(\frac{\pi}{4} \sin 0 - 0 \right) + \pi \int_0^{\frac{\pi}{2}} \frac{\sin 2x}{2} dx \\
 &= \frac{\pi^3}{8} - \frac{\pi}{4} \left[\cos 2x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{\pi^3}{8} - \frac{\pi}{4} (\cos \pi - \cos 0) \\
 &= \frac{\pi^3}{8} - \frac{\pi}{4} (-1 - 1) \\
 &= \frac{\pi^3}{8} + \frac{\pi}{2}
 \end{aligned}$$

$$2(c) \int_{-1}^1 \cos x (e^x - e^{-x}) dx$$

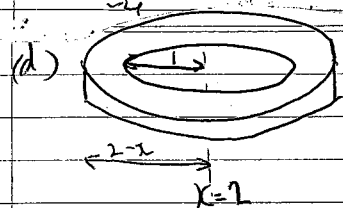
$$\text{let } f(x) = \cos x (e^x - e^{-x})$$

$$f(-x) = \cos(-x) (e^{-x} - e^{+x})$$

$$f(x) = \cos x (e^x - e^{-x}) \quad (\cos x \text{ is even})$$

$f(x)$ is odd

$$\int_{-1}^1 \cos x (e^x - e^{-x}) dx = 0$$



$$(i) A = \pi [(2-x)^2 - 1^2]$$

$$= \pi (4 - 4x + x^2 - 1)$$

$$A = \pi (3 - 4x + x^2)$$

$$(ii) \text{Volume of a slice,}$$

$$\Delta V = \pi (3 - 4x + x^2) \Delta y$$

$$= \pi (3 - 4\sqrt{y} + y) \Delta y$$

$$\text{Total volume} = \lim_{\Delta y \rightarrow 0} \sum_{j=0}^3 \pi (3 - 4\sqrt{y_j} + y_j) \Delta y$$

$$= \pi \int_0^1 (3 - 4\sqrt{y} + y) dy$$

$$= \pi \left[3y - \frac{4y^{3/2}}{3/2} + \frac{y^2}{2} \right]_0^1$$

$$= \pi \left(3 - \frac{8}{3} + \frac{1}{2} \right) - (0 - 0)$$

$$\text{Volume} = \frac{5\pi}{6} \text{ units}^3$$

(3) correct value with reason

(2) states function is odd and gives value as 0.

(2) shows function is odd
(1) gives answer as 0 without reasons

(1) correct proof

(3) correct solution

(2) expresses as correct integral including having expressed as a limit

(2) correct answer without having expressed as a limit

(1) finds volume of a slice in terms of y

$$3(a) \int x^3 \sin x dx$$

$$= \int x^3 \cdot \sin x dx \quad u = x^3 \quad v' = \sin x$$

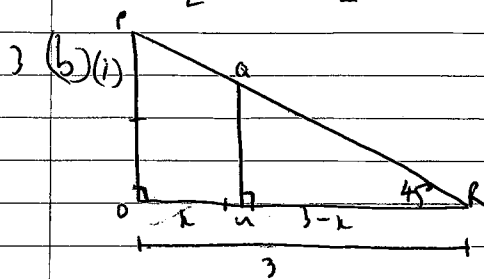
$$= -\frac{x^3}{3} \cos x + \int x \cos x dx \quad u' = 2x \quad v = -\frac{1}{2} \cos x$$

$$= -\frac{x^3}{3} \cos x + \frac{1}{2} \sin x + C$$

(3) correct answer

(2) correctly splits integral and uses integration by parts

(1) correctly splits and attempts to integrate by parts



$$\tan 45^\circ = \frac{Ox}{x}$$

$$Ox = xR$$

$$= 3 - x$$

$$\therefore \text{Area of } \triangle QST = \frac{xy(3-x)}{2}$$

$$= \frac{y(3-x)}{2}$$

$$= \frac{\sqrt{9-x^2}(3-x)}{2}$$

$$\therefore A = \int_0^3 \frac{(3-x)(9-x^2)^{1/2}}{2} dx$$

$$(ii) \text{Volume of slice } \Delta V = \frac{(3-x)(9-x^2)^{1/2}}{2} \Delta x$$

$$\text{Volume } V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 \frac{(3-x)(9-x^2)^{1/2}}{2} \Delta x$$

$$= \int_0^3 \frac{(3-x)(9-x^2)^{1/2}}{2} dx$$

$$= \frac{3}{2} \int_0^3 \sqrt{9-x^2} dx + \int_0^3 -\frac{x}{2} (9-x^2)^{1/2} dx$$

$$= \frac{3}{2} \pi \times \frac{3^2}{4} + \frac{1}{2} \int_0^3 -2x (9-x^2)^{1/2} dx$$

$$= \frac{27\pi}{4} + \frac{1}{2} \left[\frac{2}{3} (9-x^2)^{3/2} \right]_0^3$$

$$= \frac{27\pi}{4} + \frac{1}{3} (9-x^2)^{3/2}$$

(3) correct solution

(1) finds $\cos = 3-x$

(1) expresses ST as $\sqrt{9-x^2}$

(1) attempts to use area of triangle.

(2) correct answer

(2) splits integral and evaluates one correctly.

(2) evaluates answer without showing desired result, expressing as a limit.

(1) expresses volume as a limit

$$= \frac{27\pi}{4} + \frac{1}{3} (0 - 9^{\frac{3}{2}})$$

$$= \frac{27\pi}{4} - 9$$

$$= \frac{1}{4} (27\pi - 36) \text{ cm}^3 \text{ as reqd.}$$

(c) let $I_n = \int \frac{x^n}{1+x^2} dx$

(i) $I_n = \int \frac{x^{n-2} \cdot x^2}{1+x^2} dx$
 $= \int x^{n-2} \left(\frac{1+x^2 - 1}{1+x^2} \right) dx$ ✓

(2) correct solution
 (1) splits integral and manipulates algebraically

$$= \int x^{n-2} dx - \int \frac{x^{n-2}}{1+x^2} dx$$

$$I_n = \frac{x^{n-1}}{n-1} - I_{n-2}$$
 ✓

(ii) $I_5 = \frac{x^4}{4} - I_3$

$$= \frac{x^4}{4} - \left(\frac{x^2}{2} - I_1 \right)$$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} \right] + \int \frac{x}{1+x^2} dx$$

$$= \left(\frac{81}{4} - \frac{9}{2} \right) - \left(\frac{1}{4} - \frac{1}{2} \right) + \frac{1}{2} \left[\ln(1+x^2) \right]_1^3$$

$$= 20 - 4 + \frac{1}{2} (\ln 10 - \ln 2)$$

$$= 16 + \frac{1}{2} \ln 5$$
 ✓

(3) correct answer

(2) applies recurrence formula and integrates

(1) expresses I_5 in terms of I_3